

1. The `cin` Object

Topic: Reading input

Task: Write a program that asks the user to enter their first name and age, then prints:
Hello, [name]! You are [age] years old.

2. Mathematical Expressions

Topic: Arithmetic operators

Task: Write a program that reads two integers from the user. Compute and display:

- Sum
- Difference
- Product
- Integer quotient
- Remainder

3. Type Conversion (Implicit)

Topic: Implicit type conversion / mixed expressions

Task: Read an `int` and a `double`. Add them and store the result in:

- an `int` variable
- a `double` variable

Print both results. Observe and explain any difference.

4. Overflow and Underflow

Topic: Integer limits, overflow/underflow

Task: Include `<climits>`. Print the maximum `int` value. Add 1 to that maximum and print the result. Then print the minimum `int` value and subtract 1 from it. Explain what happens.

5. Type Casting

Topic: Explicit casting (`static_cast`)

Task: Read two integers. Use `static_cast<double>` to compute their precise average. Display the average with 1 decimal digit.

6. Multiple Assignment and Combined Assignment

Topic: `=` chaining, `+=`, `-=`, `*=`, etc.

Task: Declare three `int` variables. Assign them all the value `50` in one statement. Then:

- Add `10` to the first using `+=`
 - Multiply the second by `2` using `*=`
- Print all three values after the modifications.

7. Formatting Output

Topic: `<iomanip>`, `setw`, `fixed`, `setprecision`

Task: Display a table for numbers 1 through 5 with columns:

- Number (width 10, right aligned)
 - Square (width 10, right aligned)
 - Square root (width 10, 3 decimal places, right aligned)
- Add column headers. Make the output neat and aligned.

8. Working with Characters and string Objects

Topic: char, string, getline

Task: Ask the user for their full name (allow spaces, use `getline`) and their favourite letter. Print:

Hello, [full name]! Your favourite letter is [letter].

Also output the ASCII code of that letter using `int` conversion.

9. More Mathematical Library Functions

Topic: `<cmath>`, `abs`, `sqrt`, `pow`, `sin`

Task: Read a floating-point number. Display:

- Its absolute value
 - Its square root
 - Its cube (use `pow`)
 - Its sine (input is in radians)
- Include the `<cmath>` header.
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10. Hand Tracing a Program (Focus on Debugging)

Topic: Variable tracing, pre/post increment/decrement

Task:

Predict the final values of `x`, `y`, and `z` by hand-tracing the code below. Then write a program to check your prediction.

cpp

```
int x = 5, y = 10, z;
```

```
z = x++ * --y;
```

```
x += y;
```

```
y = z / x;
```

Show the step-by-step values you expect before running the program.

11. Problem Solving: A Case Study

Topic: Putting it all together (input, math, formatting)

Task: Create a simple cash register program.

- The user enters the price and quantity of **two** items.
 - Compute the subtotal.
 - Apply a sales tax of **7.5%**.
 - Display a receipt showing each item's total, the subtotal, tax amount, and final total.
- Format all currency values to two decimal places.